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Maximum a posteriori detector for linear and nonlinear channel impairment

Abstract

A receiver configured to receive a plurality of symbols is disclosed. The receiver includes a hard decision decoder, a look-up table (LUT) coupled to the hard decision decoder, and a soft metric generator coupled to the LUT. The hard decision decoder is to receive a first set of symbols from the plurality of symbols and provide a set of hard coded neighboring symbols to the LUT. The first set of symbols comprises a center symbol with neighboring symbols. The LUT is to store a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols. The soft metric generator is to calculate bit log likelihood ratio (LLR) values based on the center symbol and the value representative of the center symbol stored in the LUT.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims benefit under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/576,443, filed Mar. 16, 2023, entitled “MAXIMUM A POSTERIORI DETECTOR FOR LINEAR AND NONLINEAR CHANNEL IMPAIRMENT,” the entire disclosure of which is hereby incorporated by reference herein.

FIELD

(1) This disclosure relates generally to the field of photonic systems and more particularly relates to a maximum a posteriori (MAP) detector for both linear and nonlinear channel impairment.

BACKGROUND

(2) Channel inter-symbol interference (ISI) due to narrow filtering or nonlinear distortion in a transmitter cannot be easily compensated by a linear equalizer. The transfer function of a linear equalizer can be optimized to minimize mean square error for a narrow-filtered signal. The linear equalizer generates enhanced noise at high frequency when equalizing narrow-filtered signal. A low-pass post filter can be added to suppress the enhanced noise but introduces ISI. Nonlinear distortion also can be attributed to the driver. A large voltage swing at the driver output can create a distorted constellation after a linear equalizer. Distortion can also occur in direct detection applications. The distortion can be attributed to the laser.

SUMMARY

(3) In one embodiment, this disclosure provides a receiver to receive a plurality of symbols. The receiver comprises a hard decision decoder, a look-up table (LUT) coupled to the hard decision decoder, and a soft metric generator coupled to the LUT. The hard decision decoder is to receive a first set of symbols from the plurality of symbols and provide a set of hard coded neighboring symbols to the LUT. The first set of symbols comprises a center symbol with neighboring symbols. The LUT is to

store a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols. The soft metric generator is to calculate bit log likelihood ratio (LLR) values based on the center symbol and the value representative of the center symbol stored in the LUT.

(4) In another embodiment, the receiver includes a post filter to receive the first set of symbols from the plurality of symbols and provide a second set of symbols to the soft metric generator. In another aspect, the post filter is to send the center symbol to the LUT to train the LUT.

(5) In another embodiment, the first set of symbols comprises N symbols, wherein N is equal to $2L+1$, wherein L is a number of symbols neighboring the center symbol on each side of the center symbol.

(6) In another embodiment, the soft metric generator is to calculate soft information of the center symbol based on the LLR values.

(7) In another embodiment, the value representative of the center symbol in the LUT comprises a first value, a second value, and a third value. The soft metric generator is to calculate the bit LLRs based on the value representative of the center symbol and the first, second, and third values in the LUT.

(8) In another embodiment, the value representative of the center symbol is transmitted from the LUT to the soft metric generator.

(9) In another embodiment, the hard decision decoder is to convert the first set of symbols from the plurality of symbols to the set of hard coded neighboring symbols by converting the set of hard coded neighboring symbols to integer representations.

(10) In one embodiment, this disclosure provides a method of operating a receiver configured to receive a plurality of symbols. The method comprises receiving, by a hard decision decoder, a first set of symbols from the plurality of symbols, wherein the first set of symbols comprises a center symbol with neighboring symbols; hard coding, by the hard decision decoder, a set of hard coded neighboring symbols; providing, by the hard decision decoder, the set of hard coded neighboring symbols to a LUT; storing, in the LUT, a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols; providing, by the LUT, a center symbol and a value representative of the center symbol stored in the LUT to a soft metric generator; and calculating, by the soft metric generator, LLR values based on the center symbol and the value representative of the center symbol received from the LUT.

(11) In another embodiment the method further comprises receiving, by a post filter, the first set of symbols from the plurality of symbols; and providing, by the post filter, a second set of symbols.

(12) In another embodiment of the method, the first set of symbols comprises N symbols, wherein N is equal to $2L+1$, and wherein L is a number of symbols neighboring the center symbol.

(13) In another embodiment the method further comprises calculating, by the soft metric generator, soft information of the center symbol based on the LLR values of the value representative of the center symbol.

(14) In another embodiment of the method, the value representative of the center symbol in the LUT comprises a first value, a second value, and a third value, wherein each of the first value, second value, and third value is representative of the center symbol. The method further comprises calculating, by the soft metric generator, bit LLRs based on the value representative of the center symbol and the first, second and third values in the LUT.

(15) In another embodiment the method further comprises training the LUT. Training the LUT comprises hard coding, by the hard decision decoder, the set of hard coded neighboring symbols to address the center symbol; selecting, by the soft metric generator, all candidates of the center symbol addressed by the set of hard coded neighboring symbols; calculating, by the soft metric generator, multiple values of the candidates of the center symbol based on average values of the candidates of the center symbols; and storing, by the soft metric generator, the multiple values of the candidates of the center symbol in the LUT; calculating, by the soft metric generator, soft information of the center symbol as bit LLR values based on the multiple values of the candidates of the center symbol stored in the LUT.

(16) In another embodiment, the method further comprises converting, by the hard decision decoder, the first set of symbols from the plurality of symbols to the set of hard coded neighboring symbols by converting the set of hard coded neighboring symbols to integer representations.

(17) In one embodiment, the present disclosure provides a system for processing a plurality of symbols. The system comprises an optical hybrid circuit, a hard decision decoder coupled to the optical hybrid circuit, a LUT coupled to the hard decision decoder, and a soft metric generator coupled to the LUT. The optical hybrid circuit is to receive an optical signal and convert the optical signal to an electrical signal. The hard decision decoder is to receive a first set of symbols from the plurality of symbols. The first set of symbols comprises a center symbol with neighboring symbols and provide a set of hard coded neighboring symbols to the LUT. The LUT is to store a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols. The soft metric generator is to calculate bit LLR values based on the center symbol and the value representative of the center symbol stored in the LUT.

(18) In another embodiment, the system comprises a forward error correction (FEC) decoder coupled to the soft metric generator to receive a soft metric from the soft metric generator.

(19) In another embodiment, the system comprises an optical to electrical converter circuit coupled to the optical hybrid circuit to convert the optical signal to an electrical signal and an analog to digital converter coupled to the optical to electrical converter circuit to convert the electrical signal to a digital signal defining a plurality of symbols based on a modulation scheme employed by a transmitter.

(20) In another embodiment, the system comprises an adaptive equalizer to adapt to time varying properties of an optical channel coupled to the optical hybrid circuit.

(21) In another embodiment, the system comprises a carrier recovery circuit to estimate and compensate for frequency and phase differences between a received signal carrier wave and a local oscillator of the system.

(22) Although the disclosure relates to different aspects and embodiments, it is understood that the different aspects and embodiments disclosed herein can be integrated, combined, or used together as a combination system, or in part, as separate components, devices, and systems, as appropriate. Thus, each embodiment disclosed herein can be incorporated in each of the aspects to varying degrees as appropriate for a given implementation. Further, the various apparatus, optical elements, passivation coatings/layers, optical paths, waveguides, splitters, couplers, combiners, electro-optical devices, inputs, outputs, ports, channels, components and parts of the foregoing disclosed herein can be used comprising laser, laser-based communication system, waveguide, fiber, transmitter, transceiver, receiver, and other devices and systems without limitation. These and other features of the applicant's teachings are set forth herein.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) Unless specified otherwise, the accompanying drawings illustrate aspects of the innovations described herein. Referring to the drawings, wherein like numerals refer to like parts throughout the several views and this specification, several embodiments of presently disclosed principles are illustrated by way of example, and not by way of limitation. The drawings are not intended to be to scale. A more complete understanding of the disclosure may be realized by reference to the accompanying drawings in which:
- (2) FIG. 1 is a receiver, according to an exemplary embodiment of the disclosure.
- (3) FIG. 2 is a MAP detector portion of the receiver shown in FIG. 1, according to an exemplary embodiment of the disclosure.
- (4) FIG. 3A is a LUT for a 1D MAP soft metric calculation for a pulse amplitude modulation 2-level (PAM2) signal modulation format, according to an exemplary embodiment of the disclosure.
- (5) FIG. 3B is a simplified version of the LUT shown in FIG. 3A, according to an exemplary embodiment of the disclosure.
- (6) FIG. 4A is a LUT for a 1D MAP soft metric calculation for a pulse amplitude modulation 4-level (PAM4) signal modulation format, according to an exemplary embodiment of the disclosure. FIG. 4B is a simplified version of the LUT in FIG. 4A, according to an exemplary embodiment of the

disclosure.

(7) FIG. 5 is a linear ISI graph, according to an exemplary embodiment of the disclosure.

(8) FIG. 6 is a non-linear ISI graph, according to an exemplary embodiment of the disclosure.

(9) FIG. 7 is a logic flow diagram of a method of operating a receiver configured to receive a plurality of symbols, according to an exemplary embodiment of the disclosure.

(10) FIG. 8 illustrates a graph of the input to a PAM4 receiver, according to an aspect of this disclosure.

(11) FIG. 9 illustrates a graph of the corrected output from the PAM4 receiver, according to an aspect of this disclosure.

(12) FIG. 10 illustrates a graph of the means squared error with and without correction, according to an aspect of the disclosure.

DETAILED DESCRIPTION

(13) A sequence detector implemented either by a maximum likelihood sequence detector (MLSD) or MAP detector is necessary to correct ISI introduced by a low-pass post filter or nonlinear distortion attributed to the driver. But the complexity is prohibitively high to implement in a real-time coherent receiver. For example a MAP detector estimates transmitted symbols x given received symbol y according to Equation (1) below:

$$(14) \quad \operatorname{argmax}_x \Pr(x | y) = \operatorname{argmax}_x \frac{\Pr(y | x) \Pr(x)}{\Pr(y)} \quad (1)$$

(15) One way of MAP implementation employs a look-up table (LUT). The LUT is trained by transmitting an ensemble of transmitted symbols $\{x\}$ in according to a priori probability $\Pr(x)$ and get an averaged values of channel output as $\{D1, D2, \dots, Dm\}$, where $Dj = \langle y_j | x \rangle$, $j=1, 2, \dots, m$. Ideally, m shall be large enough to cover the channel memory. From now on, this will be referred to as mD-MAP if the LUT has m columns.

(16) If the channel function is $[h_{\text{sub.-L}}, h_{\text{sub.-L}+1}, \dots, 1, \dots, h_{\text{sub.L-1}}, h_{\text{sub.L}}]$, then $D1$ and Dm are influenced by their $2L$ neighboring transmitted symbols each, thus there are total combinations of $N_{\text{sup.}}(2L+m)$. N is the signal levels. $N=2$ for PAM2 and $N=4$ for PAM4. For example, a LUT size is $2_{\text{sup.}}9 \times 5$ with values $(D1, D2, \dots, D5)$ in each row can be created for an ISI channel according to Equation (2) below:

$$(17) \quad y(n) = x(n) + 0.2[x(n-1) + x(n+1)] + 0.1[x(n-2) + x(n+2)], x = 1, -1 \quad (2)$$

(18) A 5D-MAP LUT is shown in TABLE 1:

(19) TABLE-US-00001

TABLE 1	D1	D2	D3	D4	D5	-1.6	-1.6	-1.6	-1.4	-1.2	-1.6	-1.6	-1.4	-1.0	1.0				
-1.6	-1.4	-1.0	0.8	0.8	-1.6	-1.4	-1.0	1.0	1.4	512	{open oversize brace}	-1.4	-1.6	-1.4	-1.2	0.4	-1.4		
-1.6	-1.4	-1.2	0.6	-1.4	-1.4	-1.2	0.4	-1.0	-1.4	-1.4	-1.0	0.8	0.8	-1.4	-1.2	0.6	-0.8	0.6	
.....																			

(20) MAP requires an exhaustive search for the minimum Euclidean distance between the received signal and all the rows of the LUT, thus needs 512 calculations of distances according to Equation (3) below:

$$(21) \quad \operatorname{argmin}_D \text{.Math. } y - D \text{.Math. }^2 = \operatorname{argmin}_{\{D_1, D_2, D_3, D_4, D_5\}} \text{.Math. }^5_{k=1} \text{.Math. } y_k - D_k \text{.Math. }^2 \quad (3)$$

(22) The soft information of the center symbol can be calculated by bit LLR according to Equation (4) below:

$$(23) \quad \text{LLR}_{5\text{DMAP}}(b) = \ln \frac{\Pr(b=0|y)}{\Pr(b=1|y)} = \frac{1}{2} \min(\text{.Math. } y - D \text{.Math. }^2_{b=0}) - \frac{1}{2} \min(\text{.Math. } y - D \text{.Math. }^2_{b=1}) \quad (4)$$

To simplify the LUT size and the number of distance calculations, we create a LUT only for the center symbol $D3$, and use neighboring transmitted symbols $\{x1, x2, x4, x5\}$ as a LUT address index. The number of neighboring symbols shall be large enough to cover the channel memory.

(24) A 1D-MAP LUT is shown in TABLE 2:

(25) TABLE-US-00002

TABLE 2	x1	x2	x4	x5	D3	-1	-1	-1	-1	-1.6	-1	-1	-1	1	-1.4	-1	-1	1	-1	-1.2						
-1	-1	1	1	-1	-1	-1	-1	-1	0.4	-1	-1	-1	1	0.6	32	{open oversize brace}	-1	-1	1	-1	0.8	-1	-1	1	1	-1

1 -1 -1 -1.2 -1 1 -1 1 -1 -1 -0.8 -1 1 1 1 -0.6 -1 1 -1 -1 0.8

(26) For the same channel in Equation (2), the LUT size is reduced to from 512 to 32 since $m=1$. We call it 1D-MAP. When implementing 1D-MAP, we need first hard coded neighboring symbols: $y1.fwdarw.x1$, $y2.fwdarw.x2$, $y4.fwdarw.x4$, $y5.fwdarw.x5$ and select one from two possible candidates $D3(+)$ and $D3(-)$ as shown in Equation (5) below:

$$(27) \quad \text{argmin}_{D_3} \cdot \text{Math. } r_3 - D_3 \cdot \text{Math. } ^2 \quad (5)$$

(28) For example, $D3(+)=0.6$ and $D3(-)=-1.4$ for neighboring symbol $(-1,-1,-1,1)$ in the 1D-MAP LUT. The soft information of the center symbol can be calculated by LLR as shown in Equation (6) below:

$$(29) \quad \text{LLR}_{1\text{D-MAP}}(b) = \ln \frac{\Pr(b=0|y)}{\Pr(b=1|y)} = \frac{1}{2} \min(\cdot \text{Math. } y - D \cdot \text{Math. } ^2_{b=0}) - \frac{1}{2} \min(\cdot \text{Math. } y - D \cdot \text{Math. } ^2_{b=1}) \quad (6)$$

(30) Comparing the LLR value of 1D-MAP to the LLR value of regular symbol decision according to Equation (7):

$$(31) \quad \text{LLR}_{\text{symbol}}(b) = \ln \frac{\Pr(b=0|y)}{\Pr(b=1|y)} = \frac{1}{2} (\cdot \text{Math. } y + 1 \cdot \text{Math. } ^2) - \frac{1}{2} (\cdot \text{Math. } y - 1 \cdot \text{Math. } ^2) = \frac{2}{2} y \quad (7)$$

(32) As shown, the difference is that the boundary threshold is not fixed in 1D-MAP and depends on the neighboring symbols. Based on this proposition, the number of distance calculation can be reduced from 512 to 2. The complexity reduction can be even greater when the transmitted symbols has 4 levels.

(33) TABLE 3 shows the LUT size and number of distance calculation between 1D-MAP, 5D-MAP and maximum likelihood sequence estimation (MLSE) with a Viterbi decoder.

(34) TABLE-US-00003

TABLE 3	PAM2	PAM4	PAM2 LUT Distance	PAM4 LUT Distance	1D-MAP
2.sup.2L+1	1 × 1	2 4.sup.2L+1	1 × 1	4 5D-MAP	2.sup.2L+M × M
2.sup.2L+M	4.sup.2L+M	4 5D-MAP	2.sup.2L+M	4 5D-MAP	4.sup.2L+M × M
4.sup.2L+M	MLSE-Viterbi	2.sup.2L+1	4.sup.2L+1		

(35) In one general aspect, hard-coding neighboring symbols: $y1.fwdarw.x1'$, $y2.fwdarw.x2'$, $y4.fwdarw.x4'$, $y5.fwdarw.x5'$ according to at least one embodiment of the present disclosure significantly reduces the complexity of the design. In both linear and nonlinear ISI channels, the stored LUT value in the center symbol does not change much if there is only one symbol error in hard decision of the neighboring symbols as illustrated in TABLE 4:

(36) TABLE-US-00004

TABLE 4	x1	x2	x4	x5	D3
-1	-1	-1	-1	-1	-1.6
-1	-1	-1	-1	1	-1.4
-1	1	-1	-1	-1	-1.2
1	-1	-1	-1	-1	-1.2

(37) The MAP detector according to this disclosure significantly simplifies the complexity of a MAP detector with a small penalty. As previously discussed, MAP requires an exhaustive search for the minimum Euclidean distance between the received signal and all the rows of the LUT. in one embodiment, the MAP detector according to this disclosure provides a low power MAP detector that estimates transmitted symbols “x” given received symbol “y.” In one embodiment, the MAP detector according to this disclosure provides a MAP detector employing a simplified LUT. In one embodiment, the LUT is simplified in both size and the number of distance calculations between the received signal and all the rows of the LUT. The simplified LUT is created only for the center symbol $D3$ and neighboring transmitted symbols $\{x1, x2, x4, x5\}$ are employed as a LUT address index, where the number of neighboring symbols is large enough to cover the channel memory. In one embodiment, the LUT is trained by transmitting an ensemble of transmitted symbols $\{x\}$ according to a priori probability $\Pr(x)$ to calculate averaged values of channel output.

(38) Turning now to the figures, FIG. 1 is a receiver 100, according to an exemplary embodiment of the disclosure. The receiver 100 comprises an optical hybrid circuit 102, an adaptive equalizer 104, a carrier recovery circuit 106, a MAP detector 116, and a FEC decoder 118 to receive the soft metrics 128. The MAP detector 116 comprises a hard decision decoder 110, a soft metric generator 114, a LUT 112, and optionally a post filter 108. In one embodiment, the MAP detector 116 is a low power MAP detector.

(39) The receiver 100 receives an optical signal 101 over an optical cable from a transmitter (not

shown). The signal comprises a plurality of symbols. The optical hybrid circuit **102** comprises an optical to electrical converter circuit **103** to convert the received optical signal **101** to an electrical signal. The optical hybrid circuit **102** also comprises an analog to digital converter **105** to convert the electrical signal to a digital signal defining a plurality of symbols based on the modulation scheme employed by the transmitter. The adaptive equalizer **104** adapts to time varying properties of the optical channel. The carrier recovery circuit **106** estimates and compensates for frequency and phase differences between a received signal carrier wave and a local oscillator of the receiver. The MAP detector **116** receives the plurality of symbols, where the plurality of symbols comprises at least a first set of symbols. The first set of symbols is sent to the hard decision decoder **110**. The FEC decoder **118** linearly decodes the soft metrics **128**.

(40) FIG. **2** is a detailed view of the MAP detector **116** shown in FIG. **1** according to an exemplary embodiment of the disclosure. The MAP detector **116** receives a plurality of symbols **120** $\{y'\}$. For example, the plurality of symbols **120** are $\{y_1', y_2', y_3', \dots, y_n'\}$, indicated as “y” in FIG. **2**. The MAP detector **116** comprises a hard decision decoder **110**, a LUT **112**, and a soft metric generator **114**. In some embodiments, the MAP detector **116** further comprises a post filter **108**. Electrically connected to the MAP detector **116** is an FEC decoder **118**. In one embodiment, the MAP detector **116** is a low power MAP detector.

(41) The hard decision decoder **110** receives a first set of symbols from the plurality of symbols **120**. The first set of symbols comprises a center symbol with neighboring symbols. In one embodiment, the first set of symbols comprises N symbols, where N is equal to $2L+1$ and where L is the number of symbols neighboring the center symbol on each side of the center symbol. For example, the first set of symbols can comprise 5 symbols, where $N=5$. Thus, there are 4 neighboring symbols total, with 2 neighboring symbols on either side of the center symbol ($L=2$). For example, the first set of symbols are $\{y_1', y_2', y_3', y_4', y_5'\}$, with y_3' being the center symbol. The neighboring symbols are $\{y_1', y_2', y_4', y_5'\}$. Other values of N and L are permissible.

(42) The hard decision decoder **110** provides a set of hard coded neighboring symbols **122** $\{x\}$ to the LUT **112**. For example, the hard coded neighboring symbols are $\{x_1, x_2, x_4, x_5\}$, indicated as ‘x’ in FIG. **2**. The hard decision decoder **110** converts the first set of symbols from the plurality of symbols **120** to the set of hard coded neighboring symbols **122** $\{x\}$ by converting the neighboring symbols to integer representations. For example, the hard decision decoder receives the first set of symbols, $\{y_1', y_2', y_3', y_4', y_5'\}$. The hard decision decoder converts the neighboring symbols, $\{y_1', y_2', y_4', y_5'\}$, into integer representations, $\{x_1, x_2, x_4, x_5\}$. The integer representations are either -1 (logic 0) or 1 (logic 1) for a single bit Pam2 modulation format. The integer representations for two bits are -3 (logic 00), -1 (logic 01), 1 (logic 10), and 3 (logic 11), for a PAM4 modulation format.

(43) The hard decision decoder **110** is coupled to the LUT **112**. The LUT **112** stores at least one value representative of the center symbol that is addressable by the set of hard coded neighboring symbols **122** $\{x\}$. The hard decision decoder **110** sends the address to the LUT **112**. The LUT **112** outputs the at least one value **126** representative of the center symbol and sends the at least one value **126** to the soft metric generator **114**.

(44) In one embodiment, the soft metric generator **114** receives the at least one value **126** from the LUT **112** and calculates bit LLR values based on the center symbol and the value representative of the center symbol stored in the LUT **112**. In one embodiment, the soft metric generator **114** calculates soft information of the center symbol based on the LLR values. In one embodiment, the value representative of the center symbol in the LUT **112** comprises a first value, a second value, and a third value and the soft metric generator **114** calculates bit LLRs based on the value representative of the center symbol and the first, second and third values in the LUT **112**. In one embodiment, the at least one value **126** representative of the center symbol is transmitted from the LUT **112** to the soft metric generator **114**.

(45) With reference now to both FIGS. **1** and **2**, in one embodiment the receiver **100** comprises a post filter **108** configured to receive the first set of symbols from the plurality of symbols and provide a second set of symbols. In one embodiment, the post filter **108** filters the plurality of symbols **120** $\{y'\}$ and sends the center symbol to the LUT **112** to train the LUT **112**. Training the LUT **112** comprises

hard coding the neighboring symbols to address the center symbol, selecting all candidates of the center symbol addressed by the set of hard coded neighboring symbols **122** {x}, calculating multiple values of the candidates of the center symbol based on average values of the candidates of the center symbols, storing the multiple values of the candidates of the center symbol in the LUT, and calculating soft information of the center symbol as bit LLR values based on the multiple values of the candidates of the center symbol stored in the LUT **112**.

(46) FIG. 3A is a LUT **200** for a 1D MAP soft metric calculation for a PAM2 signal modulation format, according to an exemplary embodiment of the disclosure. In one embodiment, the LUT **200** is defined by a plurality of columns **202**, **206**, **208**, a plurality of rows **204**, and the values stored in the LUT **200**. In the embodiment illustrated in FIG. 3A, the LUT **200** comprises 2.sup.2L rows, where 2L is the channel memory. The first four columns **202** include neighboring transmitted symbols {x1,x2,x4,x5}, which are used as an address index for the LUT **200**, where each neighboring transmitted symbol {x1,x2,x4,x5} is either -1 or 1. The addresses can have more or less neighboring transmitted symbols {x1,x2,x4,x5}. The LUT **200** is addressed by the address values stored in the first four columns **202** to find the values representative of the center symbol in columns **206**, **208**, where column **206** contains the candidate center symbol D-1 for center symbol -1 and the candidate center symbol D1 for center symbol 1.

(47) With reference now to FIGS. 1 and 2 together with FIG. 3A, the LUT **200** is the LUT **112** portion of the hard decision decoder **110** of the MAP detector **116** for a one-bit scheme. The LUT **200** receives neighboring transmitted symbols {x1,x2,x4,x5} to address the values of D-1 and D1 stored in a corresponding row **204** of columns **206**, **208**. In the embodiment illustrated in FIG. 3A, for a one-bit scheme, the center symbol candidates for D-1 (-1) and D1 (1) are average values of the center symbol based on the neighboring transmitted symbol {x1,x2,x4,x5} in the first four columns **202**. The D-1 and D1 values stored in the LUT **200** are used to calculate the average according to Equation (8) to generate the LUT **210** shown in FIG. 3B for use in the least likelihood ratio calculation by the soft metric generator **114** according to Equation (9):

$$(48) \ A = \frac{D_1 + D_{-1}}{2} \quad (8) \quad \text{LLR}(b) = y - \frac{D_1 + D_{-1}}{2} \quad (9)$$

(49) Turning now to FIG. 3B, there is shown a simplified LUT **210** for a 1D MAP soft metric calculation for a PAM2 signal modulation format, according to an exemplary embodiment of the disclosure. In the embodiment illustrated in FIG. 3B, the simplified LUT **210** is a simplified version of the LUT **200** shown in FIG. 3A. The simplified LUT **210** includes a singular column **212** with rows **204** that are addressable by the neighboring transmitted symbols {x1,x2,x4,x5} in the first four columns **202**. The singular column **212** stores the average value of the center symbol calculated according to Equation (8). In one embodiment, according to Equation (8), the at least one value representative of the center symbol is now expressed as an average of the two values D-1, D1 shown in columns **206**, **208** in FIG. 3A.

(50) With reference now to FIGS. 1 and 2 together with FIGS. 3A and 3B, the LUT **200**, **210** receives the neighboring transmitted symbols {x1,x2,x4,x5} to address the average values of D-1 and D1 stored in a corresponding row **204** of columns **206**, **208** or the singular column **212** and also receives filtered symbols **124** {y} from the post filter **108** or, in another embodiment, receives unfiltered received inputs {y'} directly from the carrier recovery circuit **106** for training the LUT **210**. Training comprises hard coding the neighboring transmitted symbols {x1,x2,x4,x5} to address the candidate center symbol. The hard coded neighboring transmitted symbols {x1,x2,x4,x5} are the integer representations of the neighboring symbols. The address is the integer representations of the neighboring symbols stored in the first four columns **202**. For each address, the element transmits symbols according to a priori probability. For example, the MAP detector sends a known set of symbols to determine the actual value of the center symbol. The MAP detector calculates the value of the center symbol.

(51) Training also comprises selecting all center symbol candidates that are addressed by the set of hard coded neighboring transmitted symbols {x1,x2,x4,x5}. For example, as shown in FIG. 3B, the candidates of the center symbol for a one-bit scheme is -1 and 1. The center symbol D-1 in column

206 is the candidate for the center symbol when the center symbol is -1 . The average value of the center symbol candidate is stored in the singular column **212** is transmitted to the LUT **210** via the training input. When the LUT **210** is not being trained, the training input is optional.

(52) Training also comprises calculating multiple values of the candidates of the center symbol based on average values of the center symbol candidates. Training also comprises storing the multiple values of the center symbol candidates in the LUT **210**. Storing the values of the center symbol candidates includes storing the average values calculated according to Equation (8) for each address in rows **204**.

(53) Training also comprises calculating soft information of the center symbol as bit LLR values according to Equation (9) based on the multiple values of the candidates of the center symbol stored in the LUT **210**. In the one-bit scheme, the calculation of soft information is the subtraction of the value representative of the center symbol from the center symbol.

(54) FIG. **4A** is a LUT **300** for a 1D MAP soft metric calculation for a PAM4 signal modulation format, according to an exemplary embodiment of the disclosure. In one embodiment, the LUT **300** is defined by a plurality of columns **302**, **306**, **308**, **310**, **312**, a plurality of rows **304**, and the values stored in the LUT **300**. In the embodiment illustrated in FIG. **4A**, the LUT **300** comprises $42L$ rows, where $2L$ is the channel memory. The first four columns **302** include neighboring transmitted symbols $\{x_1, x_2, x_4, x_5\}$, which are used as an address index for the LUT **300**, where each neighboring transmitted symbol $\{x_1, x_2, x_4, x_5\}$ is either -3 , -1 , 1 , or 3 . The addresses can have more or less neighboring transmitted symbols $\{x_1, x_2, x_4, x_5\}$. The LUT **300** is addressed by the address values stored in the first four columns **302** to find the values representative of the center symbol in columns **306**, **308**, **310**, **312**, where column **306** contains the candidate center symbol $D-3$ for center symbol -3 , column **308** contains the candidate center symbol $D-1$ for center symbol -1 , column **310** contains the candidate center symbol $D1$ for center symbol 1 , and column **312** contains the candidate center symbol $D3$ for center symbol 3 .

(55) With reference now to FIGS. **1** and **2** together with FIG. **4A**, the LUT **300** is the LUT **112** portion of the hard decision decoder **110** of the MAP detector **116** for a two-bit scheme. The LUT **300** receives neighboring transmitted symbols $\{x_1, x_2, x_4, x_5\}$ to address the values of $D-3$, $D-1$, $D1$, $D3$ stored in a corresponding row **304** of columns **306**, **308**, **310**, **312**. In the embodiment illustrated in FIG. **4A**, for a two-bit scheme, the center symbol candidates for $D-3$ (-3), $D-1$ (-1), $D1$ (1), and $D3$ (3) are average values of the center symbol based on the neighboring transmitted symbol $\{x_1, x_2, x_4, x_5\}$ in the first four columns **302**. The $D-3$, $D-1$, $D1$, $D3$ values stored in the LUT **300** are used to calculate the average according to Equations (10), (11), and (12) to generate the LUT **320** shown in FIG. **4B** for use in the least likelihood ratio calculation by the soft metric generator **114** according to Equations (13)-(14):

$$(56) \quad A = \frac{D_1 + D_{-1}}{2} \quad (10) \quad B = \frac{D_1 + D_3}{2} \quad (11) \quad C = \frac{D_{-1} + D_{-3}}{2} \quad (12) \quad L(b1) = y - \frac{D_1 + D_{-1}}{2} \quad (13)$$

$$L(b0) = \min[-y + \frac{D_1 + D_3}{2}, y - \frac{D_{-1} + D_{-3}}{2}] \quad (14)$$

(57) The LUT **300** is addressed by the addresses in the first four columns **302** to find the values representative of the center symbol. As shown, for a two-bit scheme, the center symbol candidates are -3 , -1 , 1 , or 3 . The center symbol $D-3$ in column **306** is the center symbol candidate when the center symbol is -3 . The center symbol $D-1$ in column **308** is the center symbol candidate when the center symbol is -1 . The center symbol $D1$ in column **310** is the center symbol candidate when the center symbol is 1 . The center symbol $D3$ in column **312** is the center symbol candidate when the center symbol is 3 . The values stored in a row for $D-3$ is the average value of the center symbol when the center symbol is -3 given the neighboring symbols. The values stored in a row for $D-1$ is the average value of the center symbol when the center symbol is -1 given the neighboring symbols. The values stored in a row for $D1$ is the average value of the center symbol when the center symbol is 1 given the neighboring symbols. The values stored in a row for $D3$ is the average value of the center symbol when the center symbol is 3 given the neighboring symbols.

(58) Turning now to FIG. **4B**, there is shown a simplified LUT **320** for a 1D MAP soft metric calculation for a PAM4 signal modulation format, according to an exemplary embodiment of the disclosure. In the embodiment illustrated in FIG. **4B**, the simplified LUT **320** is a simplified version of

the LUT **300** shown in FIG. **4A**. The simplified LUT **320** includes three columns **322**, **324**, **326** that are addressable by addresses in rows **304**. The columns **322**, **324**, **326** store the average value of the center symbol given the neighboring transmitted symbols $\{x_1, x_2, x_4, x_5\}$ in the first four columns **302**. The value representative of the center symbol is the average value of the center symbol calculated according to Equations (10)-(12). In one embodiment, according to Equations (10)-(12), the at least one value representative of the center symbol is now expressed as an average of two values shown in columns **322**, **324**, **326** in LUT **320**. The first average value in column **322** is the average of D1 and D-1 calculated according to Equation (10). The second average value in column **324** is the average of D1 and D3 calculated according to Equation (11). The third average value in column **326** is the average of D-1 and D-3 calculated according to Equation (12).

(59) Training the LUT **320** can be done in a similar manner as training the LUT **210** described in connection with FIG. **3B**. Training the LUT **320** comprises calculating three values representative of the center symbol.

(60) Turning back to FIGS. **1** and **2**, the value representative of the center symbol is transmitted from the LUT **112** (e.g., LUTs **200**, **210**, **300**, **320**) to the soft metric generator **114**. The soft metric generator **114** calculates bit LLR values based on the center symbol and the value representative of the center symbol stored in the LUT **112**. The soft metric generator **114** calculates soft information of the center symbol based on the LLR values. The soft metric generator **114** calculates the soft information for the one-bit scheme by subtracting the value representative of the center symbol from the center symbol according to Equation (9). For the two-bit scheme, the soft metric generator **114** performs two calculations, one calculation for the first bit according to Equation (13) and a second calculation for the second bit according to Equation (14). The first bit calculation is the first value in column **322** subtracted from the center symbol. The second bit calculation is the minimum of either the second value in column **324** minus the center symbol or the center symbol minus the third value in column **326**. The soft metric generator **114** is coupled to the FEC decoder **118** to receive the soft metrics **128** from the soft metric generator **114**.

(61) In one embodiment, the MAP detector **116** comprises a post filter **108**. The post filter **108** receives a first set of symbols from a plurality of symbols **120** $\{y'\}$ and transmits a second, filtered, set of symbols **124** $\{y\}$ to the soft metric generator **114** and, if training, to the LUT **112**.

(62) FIG. **5** is a linear ISI graph **400**. The linear ISI graph **400** shows that the linear equalizer **402** provides the highest signal to noise ratio (SNR) vs bit error rate (BER). The MLSE **408** has the lowest SNR vs the BER. The graph also shows there is minimal difference between the SNR and BER between MLSE **408**, a 5D-MAP **406**, and a 1D-MAP **404**. The 1D-MAP **404** is most align with the disclosure herein.

(63) FIG. **6** is a non-linear ISI graph **410**. The non-linear ISI graph **410** shows that the 1D-MAP **414** and the 5D-MAP **416** have almost an identical optical SNR (OSNR) vs BER. The 1D-MAP **414** and 5D-MAP **416** also have a better ratio than when no compensation **412** is done. The linear driver **418** provides the best ratio of OSNR vs BER.

(64) FIG. **7** is a logic flow diagram of a method **500** of operating a receiver configured to receive a plurality of symbols, according to an exemplary embodiment of the disclosure. With reference to FIGS. **1** and **2** together with FIG. **7**, in one embodiment a hard decision decoder **110** receives **502** a first set of symbols from the plurality of symbols **120**. The first set of symbols comprises a center symbol with neighboring symbols. In one embodiment, the first set of symbols comprises N symbols, wherein N is equal to $2L+1$, and wherein L is a number of symbols neighboring the center symbol. The hard decision decoder **110** hard code **504** a set of hard coded neighboring symbols. The hard decision decoder **110** provides **506** the set of hard coded neighboring symbols to a LUT **112**. The LUT **112** stores **508** a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols. The LUT **112** provides **510** a center symbol and a value representative of the center symbol stored in the LUT **112** to a soft metric generator **114**. The soft metric generator **114** calculates bit LLR values based on the center symbol and the value representative of the center symbol received from the LUT **112**.

(65) In one embodiment, according to the method **500** the hard decision decoder **110** converts the first

set of symbols from the plurality of symbols **120** {y'} to the set of hard coded neighboring symbols **122** {x} by converting the hard coded neighboring symbols **122** {x} to integer representations.

(66) In one embodiment, according to the method **500**, the post filter **108** receives the first set of symbols from the plurality of symbols **120** (y') and provides a second set of symbols **124** {y} to the soft metric generator **114** and, in some embodiments, provides the second set of symbols **124** {y} to the LUT **112**. In one embodiment, according to the method **500**, the soft metric generator **114** calculates soft information of the center symbol based on the LLR values of the value representative of the center symbol received from the LUT **112** and the second set of symbols **124** {y} received from the post filter **108**.

(67) In one embodiment, the value representative of the center symbol in the LUT **112** comprises a first value, a second value, and a third value representative of the center symbol. According to the method **500**, the soft metric generator **114** calculates bit LLRs based on the value representative of the center symbol and the first, second and third values in the LUT.

(68) In one embodiment, the method **500** comprises training the LUT **112**. According to the method **500**, the hard decision decoder **110** generates the hard coded neighboring symbols **122** {x} to address the center symbol in the LUT **112**. The soft metric generator **114** selects candidates of the center symbol addressed by the set of hard coded neighboring symbols **122** {x}. The soft metric generator **114** calculates multiple values of the candidates of the center symbol based on average values of the candidates of the center symbols. The soft metric generator **114** stores the multiple values of the candidates of the center symbol in the LUT **112**. The soft metric generator **114** calculates the soft information of the center symbol as bit LLR values based on the multiple values of the candidates of the center symbol stored in the LUT **112**.

(69) In one embodiment, the method of FIG. 7 and the hard decision decoder of FIGS. 1 and 2 can be used for direct detection applications such as direct detection receivers. For example, direct detection receivers may be a PAM4 receiver. The disclosed method can be used for nonlinearity compensation in direct detection. The method can correct data dependent non-linearity introduced in the transmitter by the laser.

(70) In general, the method includes making a tentative decision of pre/post symbols to create four correction terms on the current symbol, one term for each potential decision of PAM4. The correction terms are adaptive and converge over time using the adaptation logic. The method calculates slicer error for every PAM4 decision based on the corresponding correction term. The output is the symbol with minimum slicer error after the non-linear (NL) correction term.

(71) FIG. 8 illustrates a graph of the input to a PAM4 receiver, according to an aspect of this disclosure. The PAM4 graph **610** illustrates the four levels, D-3 (-3) **612**, D-1 (-1) **614**, D1 (1) **616**, and D3 (3) **618**. As shown, the four levels -3, -1, 1, 3 are distorted and not in line. The distortion is caused by the nonlinearity of the laser.

(72) FIG. 9 illustrates a graph **630** of the corrected output from the PAM4 receiver, according to an aspect of this disclosure. As shown, there is no distortion to the levels, D-3 (-3) **632**, D-1 (-1) **634**, D1 (1) **636**, and D3 (3) **638**. The levels are in line. The disclosed method remedies the distortion at the input to the receiver by outputting non-distorted signals.

(73) FIG. 10 illustrates a graph of the means squared error with and without correction, according to an aspect of the disclosure. The graph **700** shows improvement of MSE with correction, from the uncorrected MSE **702** to the MSE with non-linear correction (NLC) adaptation using all symbols per clock cycle **706**. The graph **700** shows 2 dB of improvement from the uncorrected MSE **702** to the MSE with NLC adaptation on all symbols **706**. The MSE without correction **702** has a higher MSE. The means squared error with NLC adaption on all chunk symbols **706** has lowest MSE. To save power consumption, we can use only 1 symbol per clock cycle to adaptively train LUT. The means squared error with NLC adaptation on one symbol per clock cycle **704** converges to the MSE with NLC adaptation on all symbols **706**.

(74) Having thus described several aspects and embodiments of the technology of this application, it is to be appreciated that various alterations, modifications, and improvements will readily occur to those of ordinary skill in the art. Such alterations, modifications, and improvements are intended to be within

the scope of the technology described in the application. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described. In addition, any combination of two or more features, systems, articles, materials, and/or methods described herein, if such features, systems, articles, materials, and/or methods are not mutually inconsistent, is included within the scope of the present disclosure.

(75) In most embodiments, a processor may be a physical or virtual processor. In other embodiments, a virtual processor may be spread across one or more portions of one or more physical processors. In certain embodiments, one or more of the embodiments described herein may be embodied in hardware such as a Digital Signal Processor (DSP). In certain embodiments, one or more of the embodiments herein may be executed on a DSP. One or more of the embodiments herein may be programmed into a DSP. In some embodiments, a DSP may have one or more processors and one or more memories. In certain embodiments, a DSP may have one or more computer readable storages. In many embodiments, a DSP may be a custom designed ASIC chip. In other embodiments, one or more of the embodiments stored on a computer readable medium may be loaded into a processor and executed.

(76) Also, as described, some aspects may be embodied as one or more methods. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

(77) The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases.

(78) As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified.

(79) The terms “approximately” and “about” may be used to mean within $\pm 20\%$ of a target value in some embodiments, within $\pm 10\%$ of a target value in some embodiments, within $\pm 5\%$ of a target value in some embodiments, and yet within $\pm 2\%$ of a target value in some embodiments. The terms “approximately” and “about” may include the target value.

(80) In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. The transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively.

(81) Where a range or list of values is provided, each intervening value between the upper and lower limits of that range or list of values is individually contemplated and is encompassed within the disclosure as if each value were specifically enumerated herein. In addition, smaller ranges between and including the upper and lower limits of a given range are contemplated and encompassed within the disclosure. The listing of exemplary values or ranges is not a disclaimer of other values or ranges between and including the upper and lower limits of a given range.

(82) The use of headings and sections in the application is not meant to limit the disclosure; each section can apply to any aspect, embodiment, or feature of the disclosure. Only those claims which use the words “means for” are intended to be interpreted under 35 USC 112(f). Absent a recital of “means for” in the claims, such claims should not be construed under 35 USC 112. Limitations from the specification are not intended to be read into any claims, unless such limitations are expressly included in the claims.

(83) Embodiments disclosed herein may be embodied as a system, method or computer program

product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module,” or “system.” Furthermore, embodiments may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

Claims

1. A receiver configured to receive a plurality of symbols, the receiver comprising: a hard decision decoder; a look-up table (LUT) coupled to the hard decision decoder; and a soft metric generator coupled to the LUT; wherein the hard decision decoder is configured to: receive a first set of symbols from the plurality of symbols, wherein the first set of symbols comprises a center symbol with neighboring symbols; and provide a set of hard coded neighboring symbols to the LUT, wherein the LUT is configured to store a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols; and wherein the soft metric generator is configured to calculate bit log likelihood ratio (LLR) values based on the center symbol and the value representative of the center symbol stored in the LUT.
2. The receiver of claim 1, comprising a post filter configured to receive the first set of symbols from the plurality of symbols and provide a second set of symbols to the soft metric generator.
3. The receiver of claim 2, wherein the post filter is configured to send the center symbol to the LUT to train the LUT.
4. The receiver of claim 1, wherein the first set of symbols comprises N symbols, wherein N is equal to $2L+1$, wherein L is a number of symbols neighboring the center symbol on each side of the center symbol.
5. The receiver of claim 1, wherein the soft metric generator is configured to calculate soft information of the center symbol based on the LLR values.
6. The receiver of claim 1, wherein the value representative of the center symbol in the LUT comprises a first value, a second value, and a third value, wherein the soft metric generator is configured to calculate bit LLRs based on the value representative of the center symbol and the first, second, and third values in the LUT.
7. The receiver of claim 1, wherein the value representative of the center symbol is transmitted from the LUT to the soft metric generator.
8. The receiver of claim 1, wherein the hard decision decoder is configured to convert the first set of symbols from the plurality of symbols to the set of hard coded neighboring symbols by converting the set of hard coded neighboring symbols to integer representations.
9. A method of operating a receiver to receive a plurality of symbols, the method comprising: receiving, by a hard decision decoder, a first set of symbols from the plurality of symbols, wherein the first set of symbols comprises a center symbol with neighboring symbols; hard coding, by the hard decision decoder, a set of hard coded neighboring symbols; providing, by the hard decision decoder, the set of hard coded neighboring symbols to a look-up table (LUT); storing, in the LUT, a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols; providing, by the LUT, a center symbol and a value representative of the center symbol stored in the LUT to a soft metric generator; and calculating, by the soft metric generator, bit log likelihood ratio (LLR) values based on the center symbol and the value representative of the center symbol received from the LUT.
10. The method of claim 9, further comprising: receiving, by a post filter, the first set of symbols from the plurality of symbols; and providing, by the post filter, a second set of symbols.
11. The method of claim 9, wherein the first set of symbols comprises N symbols, wherein N is equal to $2L+1$, and wherein L is a number of symbols neighboring the center symbol.
12. The method of claim 9, further comprising calculating, by the soft metric generator, soft

information of the center symbol based on the LLR values of the value representative of the center symbol.

13. The method of claim 10, wherein the value representative of the center symbol in the LUT comprises a first value, a second value, and a third value, wherein each of the first value, second value, and third value is representative of the center symbol, and wherein the method further comprises: calculating, by the soft metric generator, bit LLRs based on the value representative of the center symbol and the first, second, and third values in the LUT.

14. The method of claim 9, further comprising training the LUT, wherein training the LUT comprises: hard coding, by the hard decision decoder, the set of hard coded neighboring symbols to address the center symbol; selecting, by the soft metric generator, all candidates of the center symbol addressed by the set of hard coded neighboring symbols; calculating, by the soft metric generator, multiple values of the candidates of the center symbol based on average values of the candidates of the center symbol; storing, by the soft metric generator, the multiple values of the candidates of the center symbol in the LUT; and calculating, by the soft metric generator, soft information of the center symbol as bit LLR values based on the multiple values of the candidates of the center symbol stored in the LUT.

15. The method of claim 9, further comprising converting, by the hard decision decoder, the first set of symbols from the plurality of symbols to the set of hard coded neighboring symbols by converting the set of hard coded neighboring symbols to integer representations.

16. A system for processing a plurality of symbols, the system comprising: an optical hybrid circuit, a hard decision decoder coupled to the optical hybrid circuit; a look-up table (LUT) coupled to the hard decision decoder; and a soft metric generator coupled to the LUT; wherein the optical hybrid circuit is configured to receive an optical signal and convert the optical signal to an electrical signal; wherein the hard decision decoder is configured to: receive a first set of symbols from the plurality of symbols, wherein the first set of symbols comprises a center symbol with neighboring symbols; and provide a set of hard coded neighboring symbols to the LUT, wherein the LUT is configured to store a value representative of the center symbol that is addressable by the set of hard coded neighboring symbols; and wherein the soft metric generator is configured to calculate bit log likelihood ratio (LLR) values based on the center symbol and the value representative of the center symbol stored in the LUT.

17. The system of claim 16, comprising a forward error correction (FEC) decoder coupled to the soft metric generator to receive a soft metric from the soft metric generator.

18. The system of claim 16, comprising: an optical to electrical converter circuit coupled to the optical hybrid circuit to convert the optical signal to an electrical signal; and an analog to digital converter coupled to the optical to electrical converter circuit to convert the electrical signal to a digital signal defining a plurality of symbols based on a modulation scheme employed by a transmitter.

19. The system of claim 16, comprising an adaptive equalizer configured to adapt to time varying properties of an optical channel coupled to the optical hybrid circuit.

20. The system of claim 19 comprising a carrier recovery circuit configured to estimate and compensate for frequency and phase differences between a received signal carrier wave and a local oscillator of the system.
